2017, a major German study of light pollution, carried out by ecologist Franz Hölker and others using satellite data, showed that the area of Earth illuminated artificially grew 9 percent between 2012 and 2016. The brightening is most intense in industrializing countries in South America, Africa, and Asia, but it also continues to increase in the already well-lit countries of Europe and in the US.

Astronomers were among the first to notice light pollution because it interfered with their ability to see celestial objects in the night sky. In 1988, American astronomers Tim Hunter and David Crawford founded the International Dark-Sky Association to protect the night skies from light pollution. It was the first organization of its kind.

Since then, studies have examined the effects of light pollution on plants and animals, which rely on the cycles of light and dark to govern life-sustaining behaviors such as nourishment, sleep, protection from predators, and even reproduction. Such research reveals a raft of ill-effects. One study showed that trees in Europe are budding more than a week earlier than they were in the 1990s; this alters their period of growth, and may mean that they fail to drop their leaves and fruit and enter the dormant phase in time to avoid damage over the winter.